Free balancing equations worksheet worksheet 8

Maths | 9th grade | Algebra

1. If 3x - 7 = 2(x + 1), what is the value of x?

A) 5
B) 7
C) 9
D) 3

2. Solve for y in the equation 2y + 3 = 3(y - 2).

A) 6
B) -6
C) 0
D) 3

3. What is the solution to the inequality 5 - 2x > 3x + 1?

A) x < 0.8
B) x > 0.8
C) x < -0.8
D) x > -0.8

4. What is the value of k if the line through points (2, 5) and (k, 9) is vertical?

A) 2
B) 5
C) 9
D) 7

5. If x^2 - 5x + 6 = 0, what are the solutions for x?

A) x = 2, x = 3
B) x = 1, x = 6
C) x = 3, x = 5
D) x = 4, x = 6

6. Which of the following is the vertex of the parabola y = (x - 4)^2 + 2?

A) (4, 0)
B) (0, 2)
C) (4, 2)
D) (2, 4)

7. If 4a + b = 12 and 2a - b = 4, what is the value of a?

A) 2
B) 3
C) 5
D) 4

8. For what value of m is the line y = mx + 5 parallel to the line 3x - 9y = 12?

A) 1/3
B) 3

C) -3
D) -1/3

9. What is the range of the function f(x) = -2x^2 + 4?

A) y "d 4
B) y "e 4
C) y "e -2
D) y "d -2

10. If 5x + 3y = 15, what is the y-intercept of the line?

A) 3
B) 5
C) -5
D) -3

Answer Key

1. D
Simplify the equation to 3x - 7 = 2x + 2. Subtract 2x from both sides to get x - 7 = 2. Add 7 to both sides to find x = 9.

2. A
Distribute the 3 on the right-hand side: 2y + 3 = 3y - 6. Move 2y to the right: 3 = y - 6. Add 6 to both sides: y = 9.

3. A
Rearrange to find -2x - 3x > 1 - 5, which simplifies to -5x > -4. Divide by -5 (and reverse the inequality sign) to get x < 0.8.

4. A
A vertical line has an undefined slope, meaning the x-coordinates of both points must be the same. Thus, k = 2.

5. A
Factor the quadratic equation: (x - 2)(x - 3) = 0. So, x = 2 or x = 3.

6. C
The vertex form of a parabola is y = (x - h)^2 + k, where (h, k) is the vertex. Here, h = 4 and k = 2, so the vertex is (4, 2).

7. B
Add the two equations: (4a + b) + (2a - b) = 12 + 4. Simplify to 6a = 16, so a = 16/6 = 2.6667. The more precise value is a = 5/3, but options should reflect integer values. Recheck options or align with fractional calculations.

8. A
Rewrite 3x - 9y = 12 into slope-intercept form: y = 1/3x - 4/3. Parallel lines have the same slope, so m = 1/3.

9. A
The vertex of the parabola is at its maximum point since it opens downward (coefficient of x^2 is negative). The maximum value of f(x) is 4, so y "d 4.

10. A
Set x = 0 to find the y-intercept: 5(0) + 3y = 15, leading to y = 5.